Activity 4.5 – Hazard Waveform Flags

1.Design Approach:

Designed a pipeline hazard waveform monitoring module using Verilog HDL to capture stall and flush signals. Implemented sequential logic to register the control signals at every positive clock edge.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Activated the stall signal.
- Tested flush signal activation.
- Checked registered outputs.
- Observed the signals in the waveform.
- Simulated using ModelSim.

3.Results:

The testbench checks stall and flush signal sampling and output generation. PASS/FAIL messages are displayed.

4.Issues Faced:

Control signals must be sampled at the correct clock edge. Reset conditions must initialize the outputs. Signal synchronization is important.

5.Conclusion:

Successfully designed and verified a pipeline hazard waveform monitoring module using Verilog HDL.